

Lethal Company Projektübergabe bis S1.39

Vollständiger technischer Handover für den nächsten ChatGPT-Chat

Lethal Company V81 - Projektübergabe - 02.09.2026

Repository: Tendas240/Lethal-Company-AI-Modding-Project

Direkter Startprompt für den neuen Chat

Nutze dieses Repository als vollständige Projektübergabe: <https://github.com/Tendas240/Lethal-Company-AI-Modding-Project>
Lies zuerst START_HERE_ChatGPT_Masterprompt.txt und anschließend die dort/README.md unter "ChatGPT - read first" genannten Dateien vollständig. Der chronologisch neueste bestätigte Stand ist verbindlich.

Aktueller kanonischer Build-/Testkandidat: S1.39 - "LC V1 S1.39 Cleanup Health Pikmin Shield". Letzter tatsächlich runtime-getesteter Referenzstand: S1.38 - "LC V1 S1.38 1440p Old Bird Resonance".

Wichtig: S1.39 ist build-/archiv-/diff-/config-verifiziert, aber noch nicht runtime-akzeptiert. Beginne mit den Tests aus Current/04_OPEN_ISSUES_AND_NEXT_TESTS.md und behandle S1.38 als letzte Runtime-Evidenz.

Aktueller Projektstand

Canonical project state: S1.39

Date: 2026-09-02

Current gameplay/test candidate: Profiles/LC V1 S1.39 Cleanup Health Pikmin Shield.r2z

Latest runtime-tested reference: Profiles/LC V1 S1.38 1440p Old Bird Resonance.r2z

Game: Lethal Company V81

Repository: <https://github.com/Tendas240/Lethal-Company-AI-Modding-Project>

S1.39 is the new canonical handover/test candidate. It has passed deterministic build, archive CRC, member-delta, Thunderstore package-manifest and configuration assertions, but it has **not yet received a runtime acceptance test**.

S1.38 is the newest profile actually run in game. It successfully loaded the S1.37 compatibility plugin, and FixCameraResolution was visually accepted by the user at 2560x1440. Mirage retention did **not** reliably carry through profile import: the user manually set neverDeleteRecordings=true in the Main Menu/LethalConfig, after which the S1.38 log confirmed the value was true. That same test phase exposed remaining natural CodeRebirth currency/map-object noise and the crane/Pikmin kill gap, which are addressed by S1.39. Earlier S1.36 runtime testing had already accepted the ship-door failsafe, complete enemies output and Pikmin microwave immunity.

Read order for ChatGPT

1. START_HERE_ChatGPT_Masterprompt.txt
2. Current/00_CURRENT_STATE.md
3. Current/01_HANDOVER_CORE.md
4. Current/02_TECHNICAL_BASELINE.md
5. Current/04_OPEN_ISSUES_AND_NEXT_TESTS.md
6. Current/05_FAILED_AND_OBSOLETE_APPROACHES.md
7. Current/06_RECENT_WORK_S1.32-S1.39.md
8. Current/03_PROJECT_CHRONOLOGY.md
9. Current/Projektstatus_S1.39.json
10. Current/Aktive_Modliste_S1.39.txt
11. Current/S1.39_BUILD_VERIFICATION.txt
12. Current/VERIFIKATION_S1.39.txt

Critical Gale import rule for S1.39

S1.39 embeds:

BepInEx/plugins/Tendas-S139CompatibilityFixes/S139CompatibilityFixes.dll

Import the .r2z with **Advanced options -> Import all files**.

If BepInEx does not log S1.39 Compatibility Fixes loaded., import:

Patches/S139CompatibilityFixes/Tendas-S139CompatibilityFixes-1.0.0.zip

into the same profile.

Do not call the S1.39 runtime safeguards tested until the marker is present.

Current build lineage

S1.29 gameplay base -> S1.30 -> S1.31 -> S1.32 -> S1.33 -> S1.34 -> S1.35 -> S1.36 -> S1.37 -> S1.38 -> S1.39

S1.29D was a diagnostic derivative only and is never a gameplay base.

Recent distinctions:

- **S1.36:** clean baseline with SCP999 disabled; source profile for deterministic later builds. Runtime accepted the local door failsafe and complete EnemyScan output; the user also confirmed Pikmins were no longer affected by CodeRebirth microwaves.
- **S1.37:** cumulative compatibility DLL adds normal natural-scrap filtering for CodeRebirth currency/credit items.
- **S1.38:** adds fixed 2560x1440 internal resolution and Lethal Resonance configured for Old Bird only. This is the latest runtime-tested reference.
- **S1.39:** disables Ogopogo/Vermin, suppresses natural Flash Turret and currency map-object generation, and adds direct CodeRebirth utility-kill protection for Pikmin/Puffmin. Health recharge is verified enabled in config.

Explicitly disabled in S1.39

Manifest: 179 entries, 173 active, 6 disabled.

- AJB-Keep_hangar_ship_door_closed
- zealsprince-Malfunxions
- Reiko88-Observer
- ProjectSCP-SCP999
- Kittenji-Dont_Touch_Me
- SoftDiamond-BrutalCompanyMinusExtraReborn

Persisting stable architecture

- Hold-to-Scan via LethalHUD.
- Pikmin water resistance.
- Company routing + auto landing via CompanyBuildingEnhancements.
- 26 intended normal interiors at Weight 100, Black Mesa through its DawnLib path.
- Rolling Giant / Shy Guy / Siren Head use native spawn ownership.
- RandomEnemiesSize active.
- x753-Mimics / CoronerMimics remain removed/disabled; fake Fire Exits must not be restored.
- Malfunxions remains off until explicit user request.
- BCMER remains off until a later isolated reactivation phase.

S1.39 immediate acceptance test

1. Import with `Import all files` and confirm S1.39 Compatibility Fixes loaded.
2. Land on a suitable moon and verify no natural Flash Turret and no normal natural CodeRebirth currency/map-object spawns.
3. Confirm Ogopogo and Vermin do not spawn.
4. Reproduce CodeRebirth Autonomous Crane proximity with Pikmin/Puffmin and confirm they cannot be killed; look for `[PikminCraneShield]` only when a blocked utility kill is attempted.
5. Damage the player and use the GeneralImprovements recharge station; verify it restores health as desired.
6. Confirm `FixCameraResolution` remains sharp/native at 2560x1440.
7. If an Old Bird appears, validate mechanical/weapon/footstep/loudspeaker replacement audio from Lethal Resonance.
8. Check `Mirage neverDeleteRecordings` after import. If it reverted, set it manually in the Main Menu/LethalConfig; profile import alone is not trusted for this per-player setting.
9. Preserve the full `LogOutput.log`.

Verbindlicher Handover-Kern

Working rules for the next ChatGPT conversation

1. Treat **S1.39** as the current canonical build/test candidate unless the repository contains a newer confirmed state.
2. Treat **S1.38** as the latest runtime-tested reference until S1.39 is actually run.
3. S1.29D is diagnostic only and must never become the gameplay base.
4. Before changing enemy spawning, identify the actual spawn owner. Prefer one positive owner per enemy.
5. Preserve the binding spawn-rate screenshot ratios for the enemies shown on Experimentation, Assurance, Vow, Offense, March, Adamance and Titan.
6. Do not modify the confirmed 26xWeight100 interior architecture without a concrete runtime reason.
7. Unknown enemy PowerLevels must not be guessed.
8. Search `05_FAILED_AND_OBSOLETE_APPROACHES.md` before reviving old solutions.
9. **Malfunxions remains disabled until the user explicitly requests reactivation.**
10. **ProjectSCP-SCP999 remains disabled.** Earlier logs proved a startup NRE.
11. **BCMER remains disabled in S1.39.** It is a later isolated project, not part of this acceptance test.

12. AJB-Keep_hangar_ship_door_closed must stay disabled while the project-local door failsafe is active.
13. CodeRebirthLib must not be installed.
14. For Gale imports containing project-local DLLs, use **Advanced options -> Import all files** and verify the plugin load marker before judging behavior.
15. Distinguish build verification from runtime acceptance. S1.39 is structurally verified, not gameplay-approved yet.

Current profile distinction

Canonical test candidate

Profiles/LC V1 S1.39 Cleanup Health Pikmin Shield.r2z

SHA-256: b510e519b4af8b683e9b9e9f4e18035f90910d2e8782f2b9e6ded5e4ecef95fe

Latest runtime-tested reference

Profiles/LC V1 S1.38 1440p Old Bird Resonance.r2z

S1.38 runtime evidence confirms the new camera-resolution plugin and S1.37 compatibility plugin loaded; the user explicitly accepted the 2560x1440 visual result. Mirage retention required manual intervention: the user set `neverDeleteRecordings=true` in the Main Menu/LethalConfig, and the subsequent S1.38 log confirmed the value. S1.38 nevertheless exposed issues now targeted by S1.39 and is therefore a reference, not the final desired endpoint.

S1.39 targeted changes

- Biodiversity OgoPogoEnabled=false.
- Biodiversity EnableVermin=false.
- Natural CodeRebirth Flash Turret generation is suppressed during `RoundManager.SpawnMapObjects`.
- Natural CodeRebirth currency map objects are suppressed during `SpawnMapObjects`; the S1.37 normal `SpawnScrapInLevel` currency filter remains.
- LethalMin CodeRebirth/Pikmin interaction toggles remain false, including `Crane.Targets.Pikmin=false` and `Crane.Squishes.Pikmin=false`.
- Because the crane still killed Pikmin in runtime despite those values, the S1.39 local plugin directly guards the CodeRebirth utility kill RPC and blocks Pikmin/Puffmin from `EnemyAI.KillEnemyOnOwnerClient` in that context.
- GeneralImprovements `AddHealthRechargeStation=true` is verified and preserved. Actual full-heal behavior still needs S1.39 runtime validation.
- S1.38's fixed 2560x1440 rendering and Old-Bird-only Lethal Resonance settings are carried forward.

Stable spawn ownership

Especially:

- Rolling Giant -> native mod configuration.
- Shy Guy / Scopophobia -> native mod configuration.
- Siren Head -> native mod configuration.

Do not force them positive through `LethalLevelLoader` again without new evidence.

Binding enemy-weight references

The seven screenshots under `References/Spawn-Rates/` are binding for relative ratios among the enemies shown. New enemies may enlarge the total pool, but reference enemies must not be independently rebalanced against one another without a deliberate new decision.

Interiors

26 normal dungeon flows rotate at Weight 100. Black Mesa uses its own `DawnLib/config` path and must not be double-registered through LLL.

Pikmin protection policy

Core settings:

- No Knock Back = true
- Invinceable Pikmin = true
- Pikmin Die In Player Death Zones = false

CodeRebirth switches must remain false:

- ACU Targets Winged Pikmin
- ACU Bullet Knockbacks Pikmin
- Crane Targets Pikmin
- Crane Squishes Pikmin
- Fan Knockbacks Pikmin
- Microwave Knockbacks Pikmin

- Flash Turret Knockbacks Pikmin
- Laser Turret Kills Pikmin
- Tornado Pulls Pikmin
- Compactor Squishes Pikmin

Confirmed historically: Flash Turret protection against direct Pikmin interaction worked. S1.39 additionally removes natural Flash Turret spawning. In the S1.36 runtime test, the user also confirmed Pikmins were no longer affected by CodeRebirth microwaves; treat microwave protection as accepted unless a later regression is observed.

Still open despite immortality/config switches:

- enemy AI wasting target selection on immortal Pikmins;
- runtime validation of the new crane kill shield;
- remaining CodeRebirth hazard edge cases.

Ship-door facts

The permanent lockout was caused by an ordinary/external close combined with AJB's unconditional power refill. The local cumulative plugin keeps power at 100% only when a living player is actually inside the landed ship, and leaves vanilla hydraulic drain intact when all living players are outside.

S1.36 runtime testing confirmed the local plugin loaded, produced DoorAudit markers, and the ship-door behavior worked as intended in the user's test. Keep AJB disabled.

EnemyScan facts

EnemyScan 1.2.1 originally filtered the terminal list to EnemyAI that had a ScanNodeProperties child. The local plugin replaces only the output-building method so all active EnemyAI with EnemyType can be listed. In S1.36 the user's ~7pm terminal screenshot was cross-checked against the runtime log and the counts matched, so the complete-list fix is accepted. It does not change spawning, PowerLevels, AI, scan nodes or bestiary data.

Mirage facts

Desired settings:

- localPlayerVolume = 0.5
- neverDeleteRecordings = true
- allowRecordVoice = true
- muteVoiceMimic = false

Storage is game-root, not BepInEx/config. The user had to set neverDeleteRecordings=true manually in the Main Menu/LethalConfig after an import; the later S1.38 log then confirmed it was true. Continue checking it after major profile imports and do not claim the .r2z alone guarantees this per-player setting.

Important disabled/parked systems

- SCP999 stays disabled.
- Malfunctions stays disabled until explicit user instruction.
- Immortal Snail maximum: 2 simultaneously.
- Observer stays disabled.
- Don't Touch Me stays disabled.
- BCMER stays disabled for S1.39; reactivation is a later isolated build.
- Peepers enemy/hazard stays removed; do not confuse with the Peeper tool.
- LethalPlaytime Boxy Boo / Huggy Wuggy / Miss Delight should not be reactivated on V81.

Unresolved Enemy PowerLevels

Do not invent values for:

- Rolling Giant
- Siren Head
- Immortal Snail
- Herobrine
- Football
- Faceless Stalker
- CodeRebirth Debt Collector / Boogey Man

Technische Baseline

Current manifest: S1.39

Profile:

Profiles/LC V1 S1.39 Cleanup Health Pikmin Shield.r2z

- 179 Thunderstore manifest entries
- 173 active
- 6 explicitly disabled
- plus one project-local cumulative compatibility plugin embedded in the archive and available separately under Patches/

Explicitly disabled:

- AJB-Keep_hangar_ship_door_closed 1.0.0
- zealsprince-Malfuctions 1.10.3
- Reiko88-Observer 2.0.1
- ProjectSCP-SCP999 2.4.0
- Kittenji-Dont_Touch_Me 1.2.8
- SoftDiamond-BrutalCompanyMinusExtraReborn 1.71.0

The exact package list is in Current/Aktive_Modliste_S1.39.txt.

S1.38 package additions carried into S1.39

- Rumi-FixCameraResolution 1.5.3
- LethalResonance-LETHALRESONANCE 4.7.8
- loaforc-loaforcsSoundAPI_LethalCompany 1.0.2

Existing compatible dependencies were intentionally kept without downgrade, including loaforc-loaforcsSoundAPI 2.0.12, ButteryStancakes-EnemySoundFixes 1.9.14, Hardy-LCMaxSoundsFix 1.2.0 and BepInExPack 5.4.2305.

Required local plugin

Patches/S139CompatibilityFixes/

Fallback package:

Patches/S139CompatibilityFixes/Tendas-S139CompatibilityFixes-1.0.0.zip

Embedded DLL:

BepInEx/plugins/Tendas-S139CompatibilityFixes/S139CompatibilityFixes.dll

Expected runtime marker:

S1.39 Compatibility Fixes loaded.

Functions

1. ship-door anti-lockout behavior;
2. DoorAudit / DoorFailsafe diagnostics;
3. complete EnemyScan terminal listing;
4. natural CodeRebirth currency/credit filtering from normal scrap generation;
5. natural CodeRebirth currency map-object filtering from indoor hazard generation;
6. natural CodeRebirth Flash Turret suppression;
7. CodeRebirth utility-kill Pikmin/Puffmin guard, intended to close the observed Autonomous Crane kill path.

Gale import requirement

Use **Advanced options -> Import all files**. If the marker is absent, import the local-mod ZIP manually.

S1.39 targeted config values

Biodiversity Ogopogo

- OgopogoEnabled = false
- EnableVermin = false

No other Biodiversity component was deliberately disabled by this change.

GeneralImprovements

- AddHealthRechargeStation = true

This value was already true in the canonical S1.36 baseline. S1.39 preserves and explicitly verifies it. Runtime full-heal acceptance remains pending.

LethalMin CodeRebirth compatibility

All of these are asserted false in the S1.39 build:

- ACU Targets Winged Pikmin
- ACU Bullet Knockbacks Pikmin
- Crane Targets Pikmin
- Crane Squishes Pikmin
- Fan Knockbacks Pikmin
- Microwave Knockbacks Pikmin

- Flash Turret Knockbacks Pikmin
- Laser Turret Kills Pikmin
- Tornado Pulls Pikmin
- Compactor Squishes Pikmin

The direct S1.39 kill guard exists because the CodeRebirth Autonomous Crane still killed a Pikmin in runtime even with the crane settings false. Separately, S1.36 runtime testing confirmed the microwave interaction no longer affected Pikmins; this is an accepted behavior unless a regression is observed.

FixCameraResolution baseline

S1.38/S1.39 intended settings:

- internal render target: 2560x1440
- Auto Size = false
- Check Resolution Every Frame = false
- HUD fixed aspect ratio: true
- HDRP Bloom/Fog/Shadow/Post Processing/Vignette: Vanilla
- Antialiasing: None
- Visor: preserved

S1.38 runtime log confirms FixCameraResolutions 1.5.3 loaded, and the user explicitly reported the result worked as intended. Keep the fixed 2560x1440 configuration unless the display setup changes.

Lethal Resonance baseline

Only these three groups are enabled:

- EnabledSounds:old_bird
- EnabledSounds:old_bird_footsteps
- EnabledSounds:old_bird_speaker

All other Lethal Resonance sound/config toggles are false.

Runtime plugin/SoundAPI loading occurred in S1.38, but an actual Old Bird encounter validating the intended replacement set is still pending.

Indoor power caps

Unchanged from S1.31. Do not rebalance while testing unrelated S1.39 fixes.

Vanilla

Moon	Indoor Power
Experimentation	4
Assurance	8
Vow	10
Embrion	12
Rend	16
Dine	16
Offense	20
Adamance	22
Artifice	22
Liquidation	22
March	24
Titan	32
Gordion / Company	0

Custom/external

Moon	Indoor Power
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EGypt	20
PsychSanctum	18
Abaddon	36
Cabal	20
Arcadia	16
Argent	18
Zenit	18
Lament	10
Flicker	14
Shutter	0
Pareidolia	36
Vigilance	18
Bozoros	32
Sanguine	20
Spectralis	22
The Iris	36
Black Mesa	28
Oxyde	not separately controllable in current export

Do not guess an Oxyde value.

26 equal-weight interiors

All intended at Weight 100 on normal moons:

1. Facility
2. Haunted Mansion
3. Mineshaft
4. Bunker
5. Drains
6. Substation
7. Liminal Facility
8. Storehouse
9. Tower
10. Circus Facility
11. Atlantean Citadel
12. Deep Sewers
13. Fractured Complex
14. Greenhouse
15. Art Gallery
16. Decrepit store
17. Expanded facility
18. Expanded Mineshaft
19. Grand Armory

20. Spooky manor
21. ScoopyCastle
22. Slaughterhouse
23. Storage Complex
24. Rubber Rooms
25. Toy Store
26. Black Mesa

Black Mesa uses its own DawnLib/config path; do not double-register through LLL.

LethalMin baseline

Core Pikmin settings:

- No Knock Back = true
- Invinceable Pikmin = true
- Pikmin Die In Player Death Zones = false

Attack Blacklist must retain the current longer list plus Leaf boy; do not replace it with older shorter handover values.

Mirage baseline

Desired/current latest-runtime-confirmed after **manual user setting**:

- localPlayerVolume=0.5
- neverDeleteRecordings=true
- allowRecordVoice=true
- muteVoiceMimic=false

The user had to set neverDeleteRecordings=true manually in the Main Menu/LethalConfig; the profile import did not reliably apply it. The later S1.38 log confirmed the manual value was active. Do not treat the embedded/profile copy as sufficient proof.

Paths:

- settings: <Lethal Company>/Mirage/settings.json
- recordings: <Lethal Company>/Mirage/Recording

Identified S1.38 enemy: The Cabinet

The user's four-legged Jester-like indoor enemy was identified as **Cabinet** from Cabinet_crew-TheCabinet 1.12.1. The S1.38 runtime log explicitly recorded Spawning Cabinet from vent followed by TheCabinet debug output. This is an identification fact, not a request to disable the enemy.

SpawnCycleFixes

Spawn Cycle Fixes 1.2.2 remains active. Do not disable it blindly if enemies feel late; inspect spawn probability/amount curves first.

Known warnings / issues not to overreact to

- SellMyScrap warnings about ShipInventoryUpdated should only be acted on if user-facing functionality is actually broken.
- InjectionLibrary warnings for Mirage/Opus native DLLs are expected scanner noise for non-.NET binaries.
- CodeRebirth can log Weather Registry unavailable while continuing to load; treat as compatibility warning unless missing weather content matters.
- NavMeshInCompany NodeHelper warnings remain known; investigate only if Company navigation is actually broken.

Offene Punkte und nächste Tests

Priority 0 - verify S1.39 local plugin import

Import S1.39 in Gale with **Advanced options -> Import all files**.

Required marker:

- S1.39 Compatibility Fixes loaded.

Also expect the cumulative EnemyScan marker.

If the S1.39 marker is absent, import Patches/S139CompatibilityFixes/Tendas-S139CompatibilityFixes-1.0.0.zip into the same profile and relaunch.

Do not judge S1.39 map-object filtering or Pikmin kill protection before this is confirmed.

Priority 1 - natural CodeRebirth currency cleanup

S1.37 filtered normal scrap rolls but S1.38 still showed NewCoinPrefab(Clone) instances through the map-object/hazard path.

S1.39 additionally filters indoor map-object generation.

Test several moons/interiors and confirm:

- no naturally generated Coin;
- no naturally generated Crisp Dollar Bill;
- no naturally generated Wayfarer's Wallet;
- no normal natural Credit Pad variants if they use the covered generation paths;
- dedicated CodeRebirth merchant/vending/enemy-drop currency mechanics still work when encountered.

When filtering activates, preserve [MapObjectFilter] log lines.

Priority 2 - Flash Turret suppression

Confirm CodeRebirth Flash Turret no longer appears as a normal natural indoor hazard.

This is spawn suppression, not merely Pikmin immunity.

Priority 3 - Ogopogo / Vermin disabled

Verify Biodiversity does not spawn Ogopogo and does not activate the related Vermin mechanic.

Config assertions passed; gameplay acceptance is pending.

Priority 4 - CodeRebirth Autonomous Crane vs Pikmin/Puffmin

S1.38 runtime showed the crane could kill Pikmin despite:

- Crane Targets Pikmin=false
- Crane Squishes Pikmin=false
- Pikmin invincibility

S1.39 adds a direct CodeRebirth utility-kill guard.

Expected:

- crane may not kill Pikmin/Puffmin;
- if CodeRebirth attempts the guarded kill path, [PikminCraneShield] should log a block;
- normal enemy kills outside that CodeRebirth utility context must not be globally disabled.

Priority 5 - health recharge station

AddHealthRechargeStation=true is already present and was explicitly verified in the profile.

Test:

1. take known damage;
2. use the ship health recharge station;
3. confirm it restores health to the desired full value and does not produce errors.

Priority 6 - 2560x1440 camera carry-forward

S1.38 loaded FixCameraResolutions successfully. Confirm S1.39 still renders sharply at the intended 2560x1440 internal target and has not regressed because of profile import behavior.

Priority 7 - Old Bird Lethal Resonance encounter validation

Plugin/config build validation exists, but a clean Old Bird encounter is still needed.

When an Old Bird appears, verify replacement audio for:

- main/mechanical/weapon group;
- footsteps;
- loudspeaker/voice group.

Also verify non-Old-Bird Lethal Resonance groups stay disabled.

Priority 8 - Mirage retention

The user had to set neverDeleteRecordings=true manually in the Main Menu/LethalConfig after profile import. The later S1.38 log confirmed that manual value at runtime.

After importing S1.39, check the setting again. If it reverted, set it manually; do not treat .r2z import as authoritative for this game-root per-player setting.

Confirmed behavior - CodeRebirth microwave vs Pikmins

S1.36 runtime testing produced an explicit user confirmation that Pikmins were no longer affected by the CodeRebirth microwaves. Treat this as accepted and **do not spend another build on it unless a later run shows a regression**.

Priority 9 - enemy target selection against immortal Pikmins

Immortal Pikmins can still waste enemy AI if enemies select them as targets. If reproducible, prefer a small target-filter compatibility patch rather than disabling enemies.

Priority 10 - exact enemy PowerLevels

Still unresolved and must not be guessed:

- Rolling Giant
- Siren Head
- Immortal Snail
- Herobrine
- Football
- Faceless Stalker
- CodeRebirth Debt Collector / Boogey Man

Use runtime, binary or asset analysis.

Later phase - BCMER reactivation

BCMER remains disabled in S1.39. Do not fold it into the S1.39 acceptance test.

After S1.39 is accepted, the logical next isolated build is a BCMER re-audit/reactivation against the accepted gameplay base, with special attention to spawn power/chance ownership.

Known warnings to leave alone unless functionality breaks

- SellMyScrap / ShipInventoryUpdated warnings.
- InjectionLibrary skipping Mirage/Opus native DLLs as non-.NET.
- CodeRebirth Weather Registry unavailable warning.
- NavMeshInCompany NodeHelper warnings.

After every meaningful run

Preserve the full LogOutput . log and compare:

- exceptions/NREs;
- plugin load list;
- S1.39 compatibility marker;
- DoorAudit / DoorFailsafe;
- EnemyScanFix;
- [MapObjectFilter] / [PikminCraneShield];
- SCP999 absence;
- Ogopogo/Vermin absence;
- currency/Flash Turret natural-spawn evidence;
- Pikmin hazard interactions;
- FixCameraResolution load/status;
- Lethal Resonance/SoundAPI load status;
- Mirage loaded settings.

Fehlgeschlagene / geparkte Ansätze

Do **not** reintroduce these without new technical evidence or explicit user instruction.

Component / approach	Status / reason
LethalCompanyVariables / EnemyRarityConfig	Historically removed; do not restore as the spawn solution.
LethalQuantities 1.2.9 as global enemy owner	Examined enemy-control sections were not the active source of truth.
AutoCompanyBuilding 1.2.1	Loaded but routing was unreliable on V81.
AutoCompanyBuilding 1.1.3	Routing could work, automatic landing did not.
RandomMoonFX 1.4.4 for Company automation	Active state interfered with manual moon selection; disabled state lacked the needed behavior.

Old Hold_Scan_Button	Replaced by working LethalHUD Hold-to-Scan.
Peepers enemy/hazard mod	Removed; do not confuse with Peeper tool.
CodeRebirthLib	Deprecated/unwanted; do not install.
ProjectSCP-SCP999	Keep disabled; S1.31-S1.34 logs proved startup NRE in SCP999.Plugin.Awake().
Gnomes	Caused V81 PlayerIsTargetable MissingMethod spam; removed.
FacilityMeltdown	Fully removed.
ASTeam Racist Hoarding Bugs sound replacer	Removed.
FearOverhauled	Removed.
LethalModDataLib	Removed after ShipWindows update; old NRE disappeared.
LethalPlaytime Boxy Boo / Huggy Wuggy / Miss Delight	Do not reactivate on V81 due to AI/collision problems.
MirageClipLimiter.dll	Unrealized plan; DLL does not exist.
Forcing Rolling Giant / Shy Guy / Siren Head positive weights through LLL	Native ownership is the reliable architecture.
AJB-Keep_hangar_ship_door_closed	Its unconditional refill can turn an ordinary close into permanent outside lockout. Replaced by narrower local failsafe.
Treating S1.34 door countdown as a failed custom algorithm	Invalid: the custom DLL never loaded; observed behavior was vanilla.
EnemyScan 1.2.1 original scan-node-filtered list as complete census	Incomplete by design; local patch replaces only list-building output.
S1.35 or S1.36 as current canonical candidate	Superseded by later builds. S1.36 remains the deterministic baseline/source profile, not the current test candidate.
S1.37 scrap-only currency filter as complete natural-currency solution	Insufficient alone: S1.38 still produced Coin through the map-object/hazard path. S1.39 adds SpawnMapObjects filtering.
Assuming LethalMin Crane Targets/Squishes Pikmin=false alone prevents all crane kills	Runtime disproved this. S1.39 adds direct CodeRebirth utility-kill protection.
Enabling all of Lethal Resonance	Not desired. Current design is Old-Bird-only.

Deliberately disabled / parked, not permanently obsolete

Malfunctions

Disabled from S1.34 onward by explicit user decision. **Do not re-enable unless the user explicitly asks.**

BCMER

SoftDiamond-BrutalCompanyMinusExtraReborn remains disabled in S1.39. It is not permanently forbidden.

If the user wants it back, do a new isolated compatibility/config audit after S1.39 acceptance. Do not let BCMER silently override established spawn power/chance ownership.

General anti-regression rule

Before proposing an old mod or configuration as a fix, search this file, 06_RECENT_WORK_S1.32-S1.39.md, and the chronology first.

Neueste Arbeitsphase S1.32-S1.39

This file preserves detailed recent handover facts so diagnosis context is not lost.

S1.32 - Leaf Boy and Mirage

Pikmins entered a multi-minute attack loop against LeafBoi(Clone). Because Pikmins are immortal, the engagement could persist indefinitely. Decision: append exactly Leaf boy to the existing current LethalMin Attack Blacklist; do not solve targeting by changing Leaf Boy spawn chance/health.

Mirage retention was changed to neverDeleteRecordings=true while preserving local volume/record/mimic settings. Mirage stores this in game-root Mirage/settings.json, which can survive/ignore profile-import expectations.

S1.32-S1.35 - ship-door lockout diagnosis

A normal/external close plus AJB's unconditional door-power refill produced permanent outside lockout. The exact original close actor remained unproven. Masked vanilla AI has no hangar-button interaction; Poltergeist ghost interaction is separate. First custom DLL in S1.33 was not imported by normal Gale profile import and therefore never ran. S1.34's hydraulic countdown was vanilla, not evidence that the algorithm failed.

S1.35 rebuilt the local compatibility plugin against V81 and added DoorAudit/DoorFailsafe plus complete EnemyScan output. Operational rule: use Gale Advanced options -> Import all files or import the standalone local-mod package.

S1.34-S1.36 - EnemyScan, Coin, Puma, SCP999

EnemyScan 1.2.1 source showed BuildEnemyCountString() filtered out EnemyAI without ScanNodeProperties. The local plugin patches the listing only.

Puma/PumaAI was identified as vanilla Feiopar.

CodeRebirth Coin was identified as its currency, collected with Denomination Analyzer/MoneyCounter and used by merchant/vending systems.

Current logs contradicted older docs by showing SCP999 2.4.0 loading and throwing a startup NRE. S1.36 disabled ProjectSCP-SCP999 and became the clean deterministic baseline.

S1.36 runtime acceptance

The user imported S1.36 with Gale Advanced options -> Import all files. Runtime confirmed the local compatibility plugin and [EnemyScanFix] marker loaded. The ship-door behavior worked as intended in the user's test. A terminal enemies screenshot taken around 7pm was cross-checked against the runtime spawn state and the listed counts matched. The user also explicitly confirmed Pikmins were no longer affected by CodeRebirth microwaves. These behaviors are accepted unless later regression evidence appears.

S1.37 - scrap-level currency filter

User wanted CodeRebirth currency to come from its dedicated systems rather than normal scrap rolls.

S137CompatibilityFixes temporarily removes these during RoundManager.SpawnScrapInLevel:

- Coin
- Crisp Dollar Bill
- Wayfarer's Wallet
- Credit Pad 100cc
- Credit Pad 500cc
- Credit Pad 1000cc

Entries are restored after the roll so dedicated CodeRebirth systems stay registered.

S1.38 - 1440p and Old Bird audio

Added FixCameraResolution 1.5.3 and configured a fixed 2560x1440 internal target. Added Lethal Resonance 4.7.8 plus the LC SoundAPI binding, with exactly three enabled groups: Old Bird, Old Bird footsteps, Old Bird speaker.

S1.38 was run and is the latest runtime-tested reference. The latest log proves:

- profile path/name was LC V1 S1.38 1440p Old Bird Resonance;
- FixCameraResolutions 1.5.3 loaded;
- S1.37 Compatibility Fixes loaded;
- [EnemyScanFix] marker present;
- SoundAPI completed its load pipeline;
- after the user manually set Mirage retention in the Main Menu/LethalConfig, Mirage loaded neverDeleteRecordings=true; the profile import itself had not reliably applied this per-player setting.

Remaining issues discovered in/around this test:

- generated NewCoinPrefab(Clone) entries still appeared through a path outside the scrap-only filter;
- the user wanted Flash Turret removed from natural hazard generation;
- Ogopogo/Vermin should be disabled;
- CodeRebirth Autonomous Crane could still kill Pikmin despite Crane Targets Pikmin=false and Crane Squishes Pikmin=false;

- Old Bird audio replacement still lacked a clean encounter validation;
- the user's four-legged Jester-like indoor enemy was identified from the runtime log as Cabinet from Cabinet_crew-TheCabinet 1.12.1, with Spawning Cabinet from vent logged at runtime.

S1.39 - broader natural map-object filter + crane shield

S1.39 cumulative plugin filters both V81 IndoorMapHazard[] and legacy SpawnableMapObject[] during RoundManager.SpawnMapObjects.

It suppresses:

- Flash Turret;
- currency/map-object identities detected as Coin, dollar bill, wallet/Wayfarer and matching currency item/scan-node names.

The S1.37 scrap filter remains separately active.

For the crane gap, S1.39 dynamically patches CodeRebirth's CodeRebirthUtils.KillEnemyOnOwnerClientRpc. While inside that CodeRebirth utility-kill context, the local plugin blocks EnemyAI.KillEnemyOnOwnerClient for Pikmin/Puffmin. This is intentionally narrower than making all Pikmins globally impossible to kill through every game system.

Config changes/verification:

- Biodiversity OgopogoEnabled=false;
- Biodiversity EnableVermin=false;
- all CodeRebirth/Pikmin toggles remain false;
- GeneralImprovements AddHealthRechargeStation=true retained;
- BCMER still disabled;
- S1.38 camera/audio settings carried forward.

S1.39 GitHub Actions run passed compilation, archive CRC, member delta, package manifests and config assertions.

Runtime acceptance is the next task.

Projektchronologie

This is a technical reconstruction of the confirmed project history, not a verbatim chat transcript.

S1.2 - profile/path-length stabilization

Long Gale profile names caused Windows/BepInEx path trouble. Permanent rule: keep profile names reasonably short.

S1.3-S1.10 - early cleanup, Pikmin and ownership work

Wild Pikmin/falloff, stamina, batteries, facility distribution and several obsolete/broken mods were cleaned up. Gnomes were identified as a V81 PlayerIsTargetable MissingMethod spam source and removed. Peepers enemy/hazard was removed. LLL was proven not to be the only spawn authority.

S1.11-S1.14 - spawn-owner isolation

A central LLL-only approach was tested and failed for some enemies. Rolling Giant, Shy Guy/Scopophobia and Siren Head returned to their native spawn configuration. Persistent rule: prefer one positive spawn owner per enemy.

S1.15-S1.23 - gameplay stabilization

Rolling Giant movement tuned, RandomEnemiesSize added, Company automation went through failed AutoCompanyBuilding/RandomMoonFX attempts, and the confirmed solution became CompanyBuildingEnhancements 2.6.0. LethalHUD Hold-to-Scan replaced old Hold_Scan_Button. Pikmin water resistance was confirmed.

S1.24-S1.28 - roster, balance and interiors

Enemy roster was restored, reference spawn-rate screenshots became binding, Immortal Snail max was set to 2, and 26 intended interiors were established at equal Weight 100. Black Mesa was integrated through its own DawnLib/config path.

S1.29 / S1.29D - CodeRebirth and Power audit

CodeRebirth 1.6.9 integrated. S1.29D was a diagnostic-only RedPill enemy-power audit derivative; it must never become a gameplay base.

S1.30 - caps, Mimicless, Pikmin shield

Mimics/CoronerMimics removed; CodeRebirth/Pikmin compatibility switches set false; indoor caps raised, then judged somewhat dense.

S1.31 - indoor power trim -4

All controllable indoor power caps reduced by 4 without changing weights/interiors/Pikmin behavior. Runtime then exposed the Leaf Boy/Pikmin endless attack loop and SCP999 startup NRE regression.

S1.32 - Leaf Boy blacklist + Mirage retention

Leaf boy appended to the current LethalMin Attack Blacklist. Mirage neverDeleteRecordings=true requested. A closed-ship-door lockout was observed; the original close trigger was not conclusively identified, but AJB's unconditional power refill caused the permanent lockout.

S1.33 - first ship-door failsafe

AJB disabled. Custom door failsafe designed to keep 100% hydraulic power only while a living player is inside and allow vanilla drain when everyone is outside. Later discovered Gale had not imported the DLL, so this version did not runtime-test the algorithm.

S1.34 - Malfunctions disabled / diagnosis

Malfunctions disabled by explicit user policy. Runtime proved S1.33 custom DLL had not loaded, confirmed EnemyScan's scan-node filter, identified Puma as vanilla Feiopar, and confirmed CodeRebirth Coin semantics. SCP999 still loaded and threw startup NRE.

S1.35 - rebuilt local compatibility plugin

V81 local plugin compiled for door anti-lockout + DoorAudit + complete EnemyScan. Standalone fallback package created. Structurally verified.

S1.36 - clean canonical baseline and first successful local-patch runtime acceptance

S1.35 plus ProjectSCP-SCP999 disabled. 176 manifest entries / 170 active / 6 disabled. This profile became the deterministic source baseline for later builds.

The subsequent runtime test confirmed the local S1.35 compatibility DLL actually loaded under Gale with `Import all files`. The user's ship-door test behaved as intended, and the enemies terminal output was cross-checked against the runtime log and matched the active enemy population. The user also confirmed Pikmins were no longer affected by CodeRebirth microwaves. These three behaviors are accepted and should not be reopened without a regression.

S1.37 - CodeRebirth natural currency scrap filter

Cumulative local plugin replaced S135 with S137. During normal RoundManager.SpawnScrapInLevel, Coin, Crisp Dollar Bill, Wayfarer's Wallet and Credit Pad 100/500/1000cc are temporarily removed from the natural scrap pool and restored afterward so dedicated CodeRebirth systems remain usable.

Build verification passed. Later runtime logs confirmed S1.37 Compatibility Fixes loaded and the EnemyScan patch marker.

S1.38 - 1440p + Old-Bird-only Lethal Resonance

Added:

- FixCameraResolution 1.5.3 with fixed 2560x1440 internal target;
- Lethal Resonance 4.7.8;
- loaforcsSoundAPI_LethalCompany 1.0.2.

Lethal Resonance was configured so only Old Bird, Old Bird footsteps and Old Bird speaker groups are enabled.

S1.38 was actually run in game and is the latest runtime-tested reference. Confirmed from the latest log:

- S1.37 compatibility plugin loaded;
- EnemyScanFix marker present;
- FixCameraResolutions 1.5.3 loaded;
- SoundAPI load pipeline completed;
- after the user manually set Mirage retention in the Main Menu/LethalConfig, the log loaded neverDeleteRecordings=true (profile import alone had not reliably applied it).

The test phase also exposed that normal generated CodeRebirth Coin objects were still appearing through a map-object/hazard path not covered by S1.37's scrap-only filter. The user also wanted Flash Turret natural spawns removed, Ogoogo/Vermin disabled, and reported the CodeRebirth Autonomous Crane could still kill Pikmin even though LethalMin's crane toggles were false. A four-legged Jester-like enemy seen indoors was identified from the same runtime log as Cabinet from Cabinet_crew-TheCabinet 1.12.1 (Spawning Cabinet from vent).

Old Bird replacement audio still lacked a clean encounter-based acceptance test.

S1.39 - cleanup + health + Pikmin kill shield

Built from the canonical S1.36 baseline while carrying forward S1.37/S1.38 behavior.

Changes:

- Biodiversity OgorongoEnabled=false;
- Biodiversity EnableVermin=false;
- natural Flash Turret suppressed during SpawnMapObjects;
- natural CodeRebirth currency map objects suppressed during SpawnMapObjects in both V81 IndoorMapHazard[] and legacy paths;
- S1.37 natural scrap currency filter retained;
- all relevant LethalMin CodeRebirth/Pikmin interaction toggles verified false;
- direct CodeRebirth utility-kill guard added for Pikmin/Puffmin, closing the observed crane-kill gap;
- GeneralImprovements AddHealthRechargeStation=true explicitly verified and preserved;
- S1.38 2560x1440 and Old-Bird-only Lethal Resonance carried forward;
- BCMER still disabled.

Build/CRC/member-delta/package/config/DLL-compilation checks all passed.

Current status: S1.39 is the canonical test candidate but is not yet runtime-accepted. S1.38 remains the latest runtime-tested reference.